Paper Review • Embodied AI / Robot Learning

ODE Policy

Robot Learning via Action Dynamics, Not Actions

Learn how actions change, not what actions are

In one sentence: A neural ODE learns the action velocity field; each action is integrated from the previous one, so execution is inherently continuous. It learns faster, generates faster and moves more smoothly, and plugs into visuomotor policies, VLA/WAM action heads and RL actors.

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Key Findings at a Glance

One change, four gains: learning, data, speed, smoothness

The policy outputs not action values but the action velocity field **$da/dt = f\theta(a, s, z)$**, integrated over physical time.

79.2%

LIBERO, 1 training epoch

Best of 6 visuomotor baselines: 43.2%

10.8×

Faster chunk generation than flow matching

Euler integration: 8 actions in only 5.7 ms

14 / 14

Highest success on every real task

Single-arm, bimanual, dexterous hand, humanoid

96.0%

As π0.5 action head, 1 epoch

Randomly initialized FM expert: 75.5%; only ~3% of params

Caveat: The largest gains occur early in training and narrow with full training: LIBERO epoch 100 93.8% vs FM-DiT 85.4%; π0.5 pretrained FM catches up by epoch 2; ImageWAM: all three heads ≈98% after epoch 6.

Current policies learn continuous motion as discrete points

1 Continuity learned implicitly

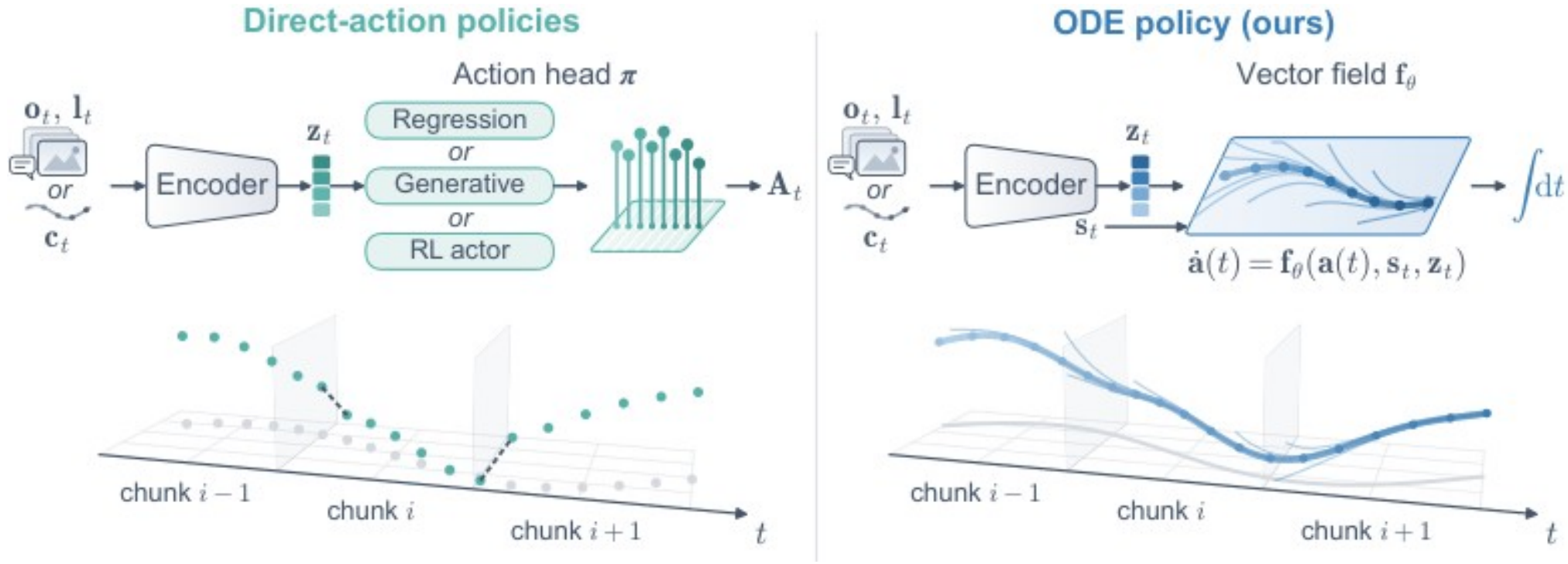
No constraint links adjacent actions; jumps or pauses often occur at chunk boundaries

2 Costly learning & inference

Needs much data and compute; diffusion/FM needs many forward passes per chunk and a full chunk before execution

3 High-freq. RL action jitter

Wastes control energy, wears motors, widens the sim-to-real gap



Core idea: policy = vector field in physical time; actions via integration

Observations
Image history + language
(RL: reference motion)

→

Encoder
Run once per chunk
→ condition z_t

→

Vector field fθ
FiLM-MLP
(a, s_t, z_t) → da/dt

↑ Proprioception s_t fed directly to field

→

Integration
RK4 / RK2 / Euler
chunk length H = 8

→

Action stream
Each action sent on arrival
to low-level controller

↺ Each action starts from the last, continuing across chunks

$$da/dt = f\theta(\, a(t), s_t, z_t \,) \qquad \hat{a}(t+k) = \hat{a}(t+k-1) + \int f\theta(\, a(t'), s_t, z_t \,)\, dt' \,, \quad k = 0 \ldots H-1$$

Physical vs. generative time

- Diffusion/FM $\tau \in [0,1]$ is generative time; intermediate samples are not reachable robot states
- Each chunk is regenerated from noise, unrelated to the previous chunk's end
- ODE t is physical time: the integration path is the executed trajectory

Training: supervise path only

- Integrate H steps from demo action a_{t−1}; L2-match to demos
- Short chunks: gradients flow through the solver, no adjoint
- No velocity labels, so differencing doesn't amplify demo noise

Inference: three benefits

- Smooth: execution is one integral curve, no jumps at chunk joins
- Streaming: send each action once integrated, no full-chunk wait
- Free step size: change step size and control rate at deployment

Early stability: contractive perturbation + seamless chunking

Contractive Perturbation

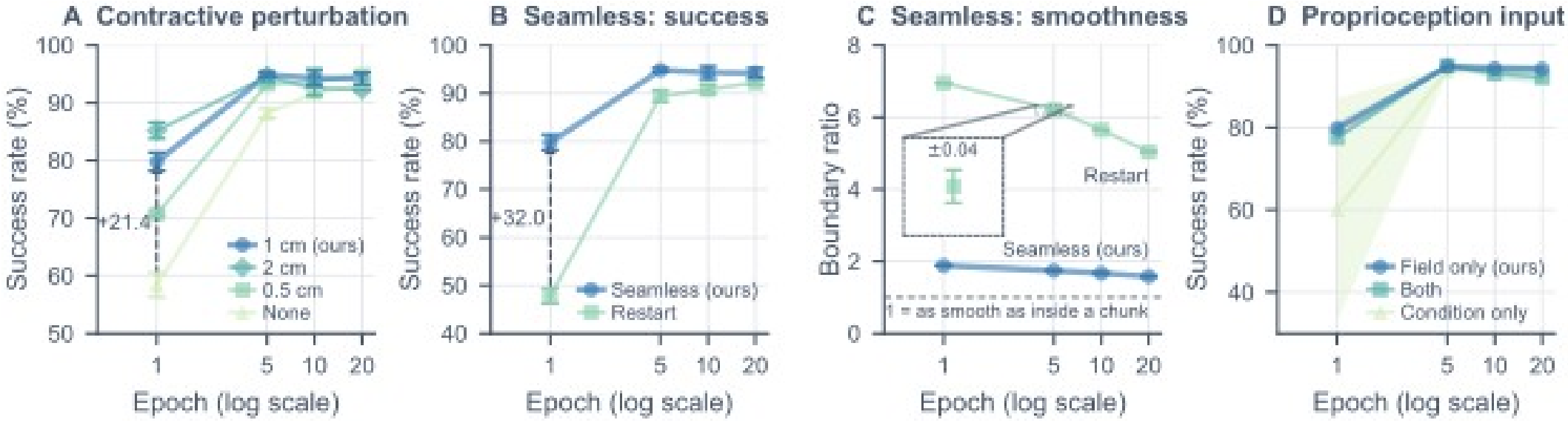
- Perturb start by δ (~1 cm / 0.03 rad, half of segments)
- Target: a_{t+k} + ρ^{k+1} • δ, ρ = 0.7; deviation decays geometrically → local contraction

Ablation: epoch 1 **79.8% → 58.4%** (removed); gap gone before epoch 20

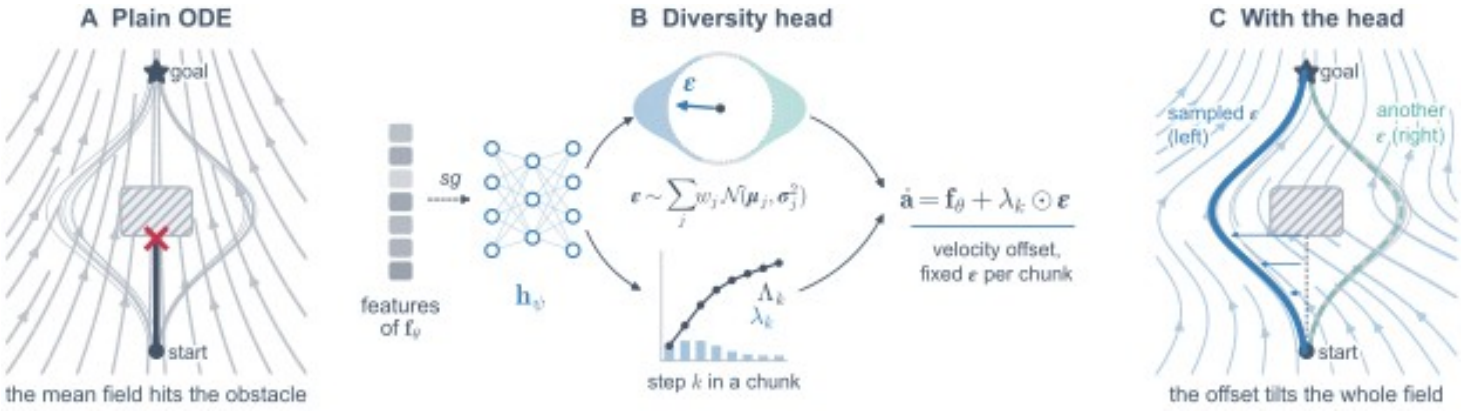
Seamless Chunking

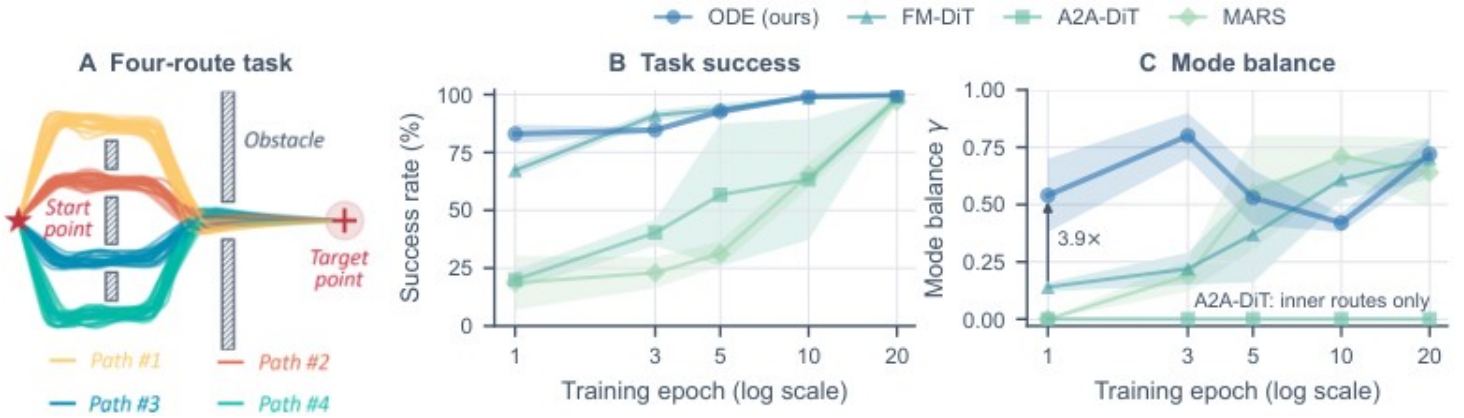
- New chunk integrates on from the previous chunk's final desired action, not restarting from measured state
- Keeps the desired action's "lead" over the measured state

Ablation: epoch 1 **79.8% → 47.8%**; at convergence 3–4× higher boundary ratio, 1.3–2.2× higher jerk



Diversity head: random tilts → multimodal deterministic ODE





Mechanism

- Velocity = fθ + λ_k ⊙ ε; ε sampled from a K=12 GMM, fixed per chunk
- Head inputs and targets are stop-gradient; fθ gradients unchanged
- Unimodal tasks (LIBERO): component mean

Results

3.9× Epoch-1 balance γ
vs. FM-DiT

- A2A-DiT takes only the inner two (γ = 0)
- FM-DiT more balanced at epoch 10; similar at epoch 20
- Real drawer: left/right arm takes apple 40% / 60%; 60% success (FM-DiT 0%)

Very fast learning: 79.2% after 1 epoch, over 85% after 2

2 vs 30

Epochs to reach 85%

A2A-DiT 4 、 FM-DiT 30 、 DDPM 50–60 、 ACT >100

87.7% > 85.4%

Data efficiency

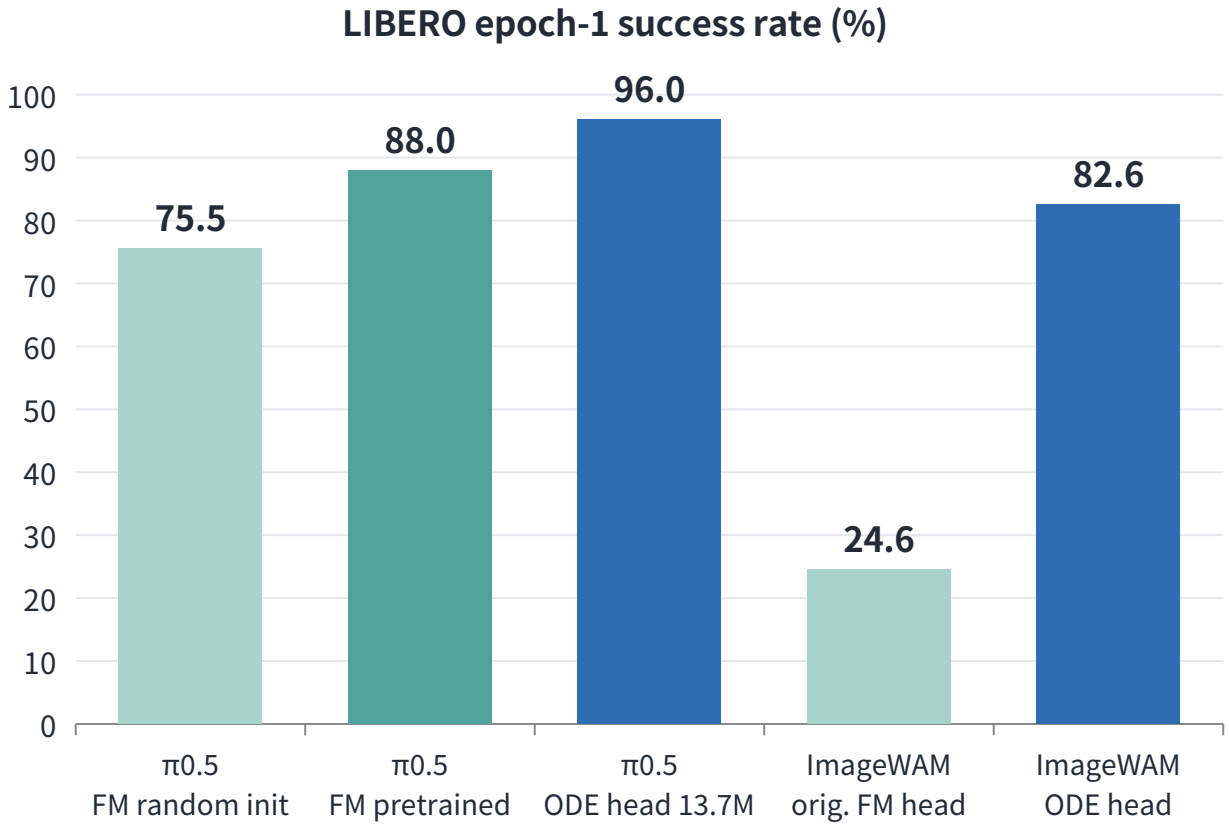
20 demos/task (epoch 100) beat FM-DiT with all 50

1.5M > 48M

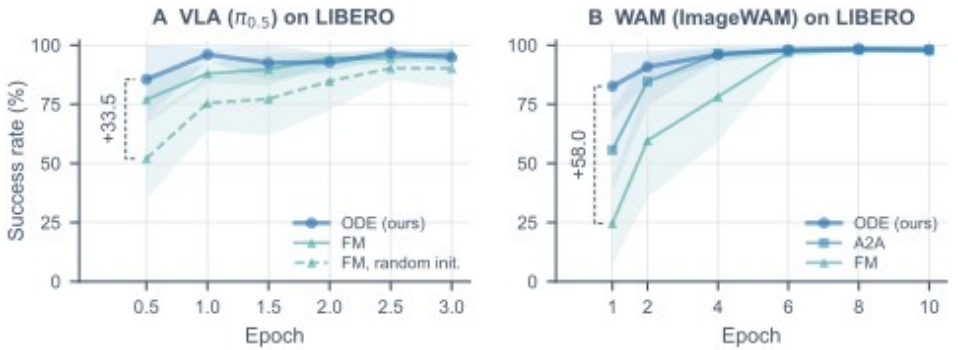
Parameter efficiency

1.5M ODE 92.5% at epoch 10; 48M FM-DiT peaks at 87.5%

Large models too: big epoch-1 leads in π0.5, ImageWAM



π0.5: single seed; ODE head has ~3% of the original expert's (430M) params, not equal capacity; ImageWAM: all three heads ≈98% after epoch 6.

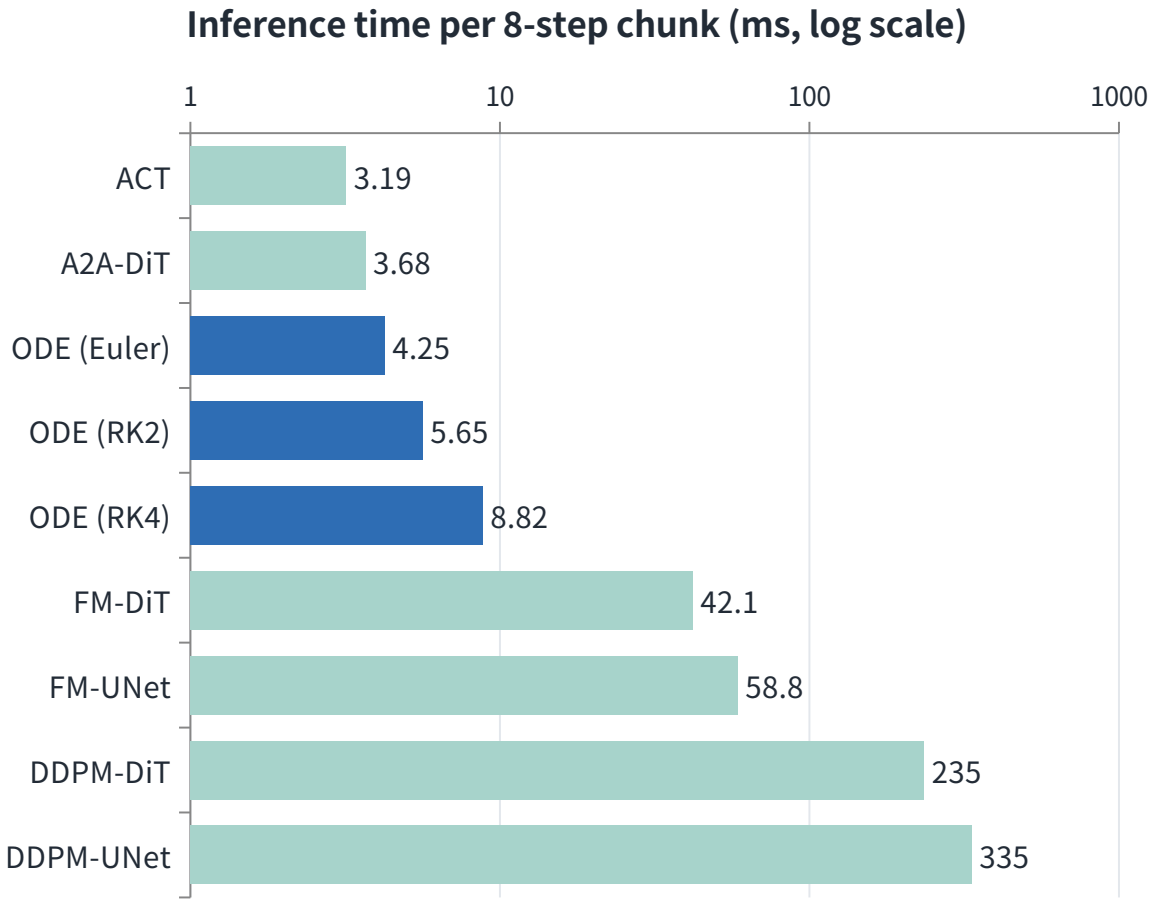


2.2 ms
π0.5 head/chunk
FM expert 31.8 ms

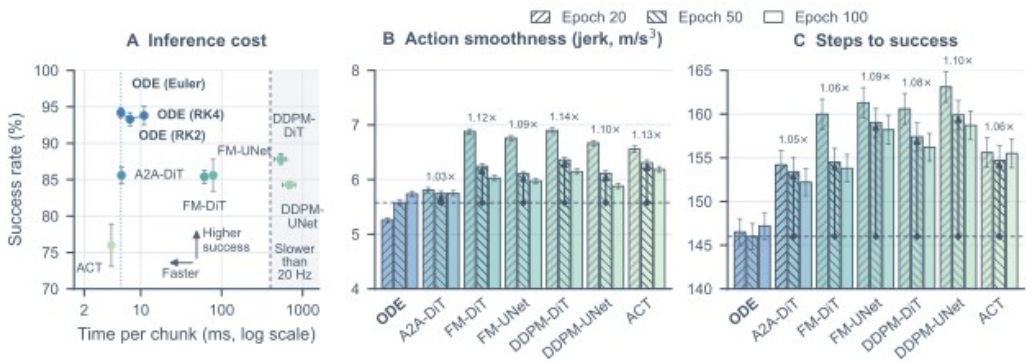
55 ms
ImageWAM model/chunk
Original FM head 223 ms

- π0.5: command change at chunk boundaries ~halved (0.097 vs 0.206)
- ImageWAM LIBERO-plus best: 89.9% vs 81.9%

Fast & good: ~single-pass speed, beats generative policies



Encoder ~1.2 ms for all methods; differences come from the head. Main text (shared GPU): Euler 5.7 ms, 1/10.8 of FM-DiT.

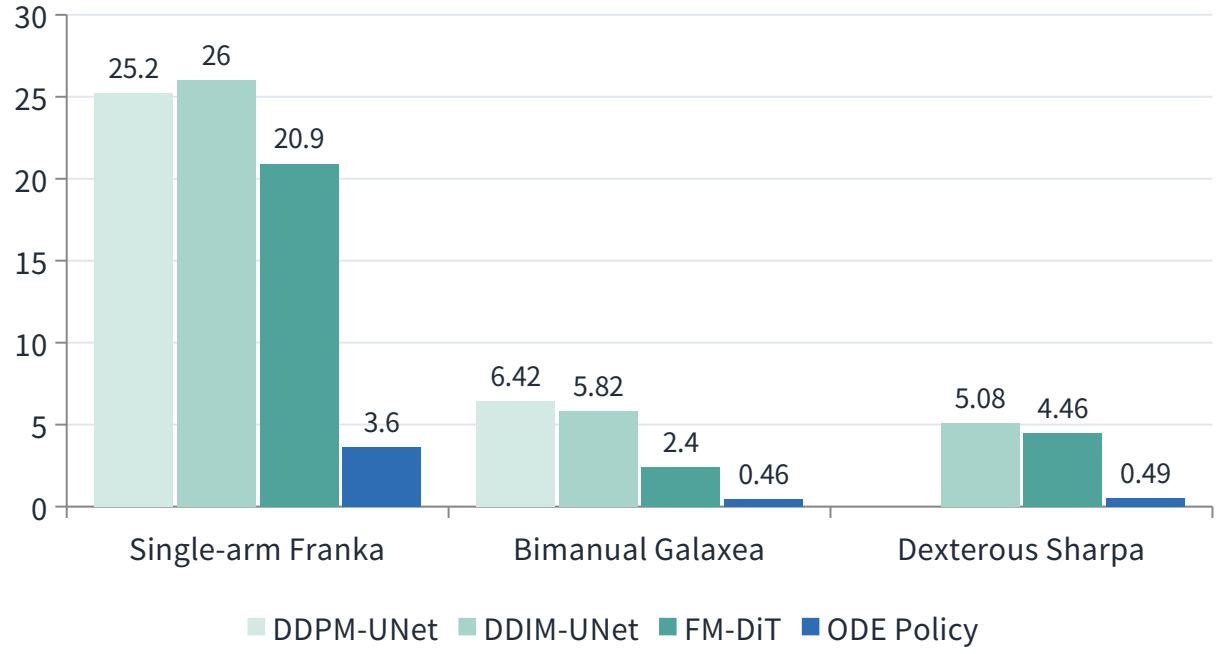


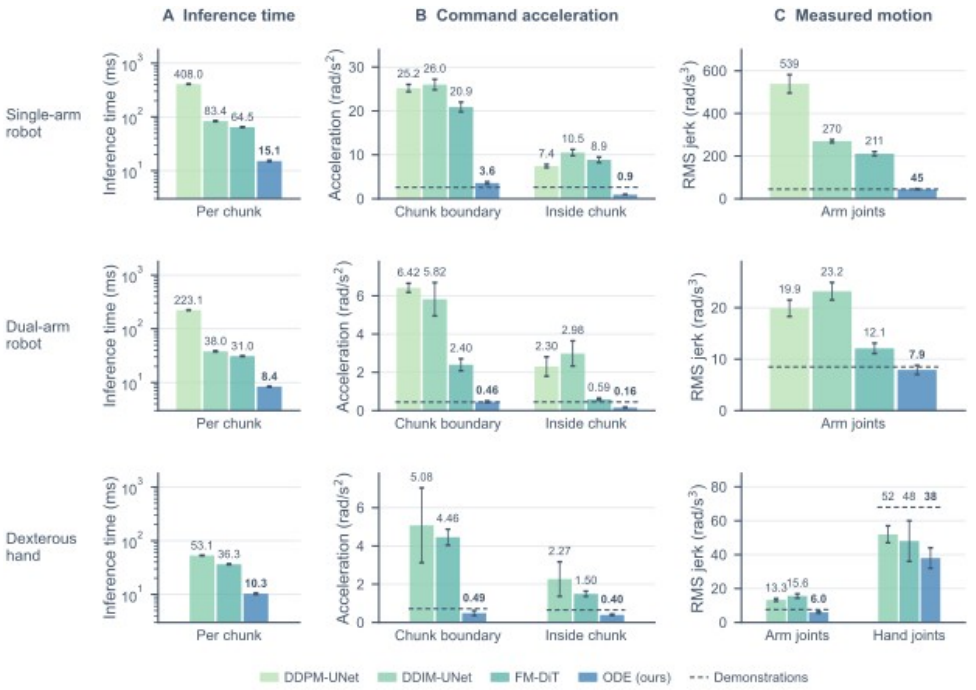
Execution quality (successful episodes)

- Lowest end-effector RMS jerk at every checkpoint; baselines 1.03–1.14×
- Fewest steps to complete (~146); baselines need 5–10% more
- RK4 / RK2 / Euler differ by < 1 pt in success → Euler suffices for deployment
- DDPM exceeds the 400 ms budget for 8 steps at 20 Hz

Real robots: smoother than human demos, ~no chunk jumps

Commanded accel. at chunk boundaries (rad/s², lower = smoother)





45 vs 211–539
1-arm jerk, rad/s³

3.5–27×
Faster than diffusion/FM

RL too: ODE actor in SuperMimic learns faster, acts steadier

+37%
Train return @0.75 h
0.630 vs 0.460

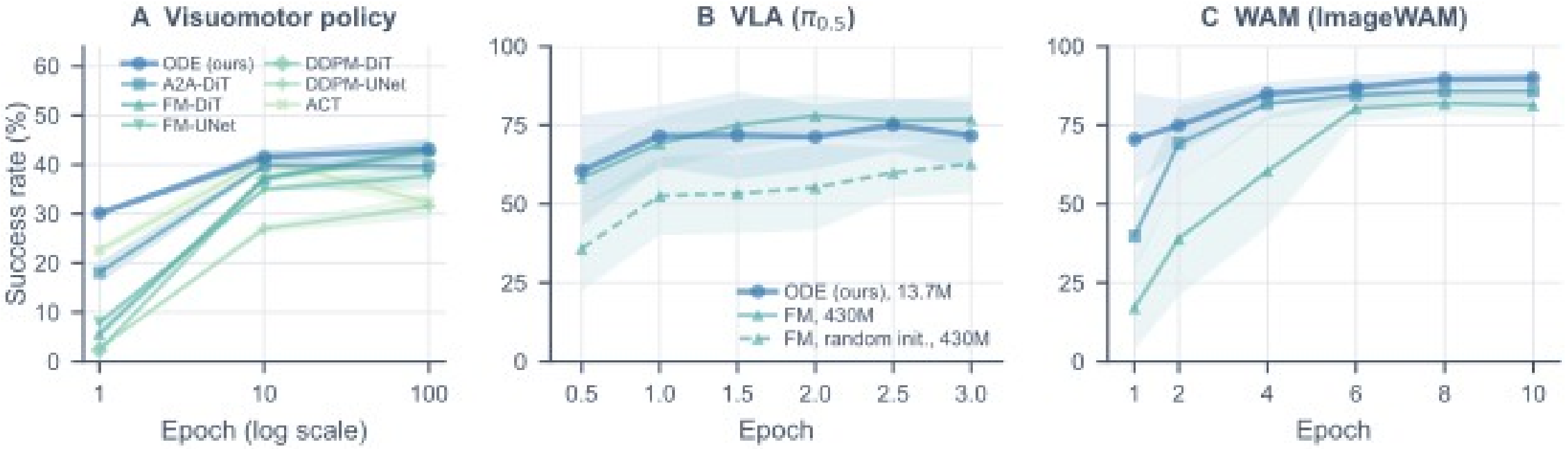
+30%
Eval. return @0.4 h
All 40 motion clips

0.885
Eval. return @3 h
Orig. actor 0.845, PPO 0.452

1.0–1.1
First/last-step grad-norm ratio
Orig. actor ~0.57

−21%
Executed accel.
vs orig. actor; < ½ PPO

Robustness: early lead; later on par with best baseline



Visuomotor policy

Epoch 1: **30.1% vs 22.6%** (ACT)

- On par with best baseline at epoch 100

ImageWAM

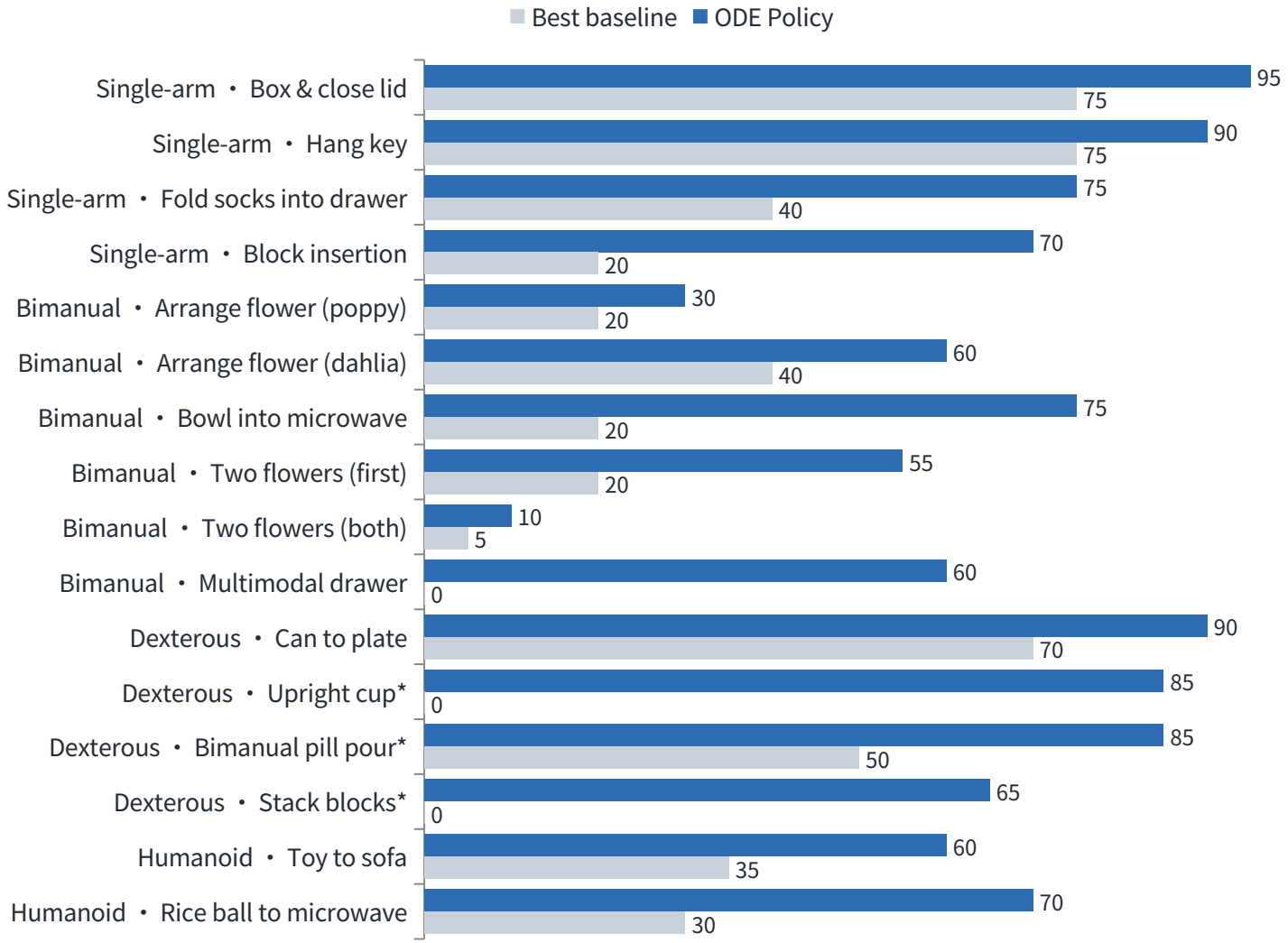
Epoch 1: **70.5% vs 17.1%**

Best checkpoint **89.9%** vs FM 81.9 / A2A 85.8

By perturbation (epoch 100)

- Largest lead on sensor noise: 34% vs ~15–22%
- Initial pose: FM/DDPM-DiT better; camera shift: all ≈ 0

14 real-robot tasks: highest success on every task

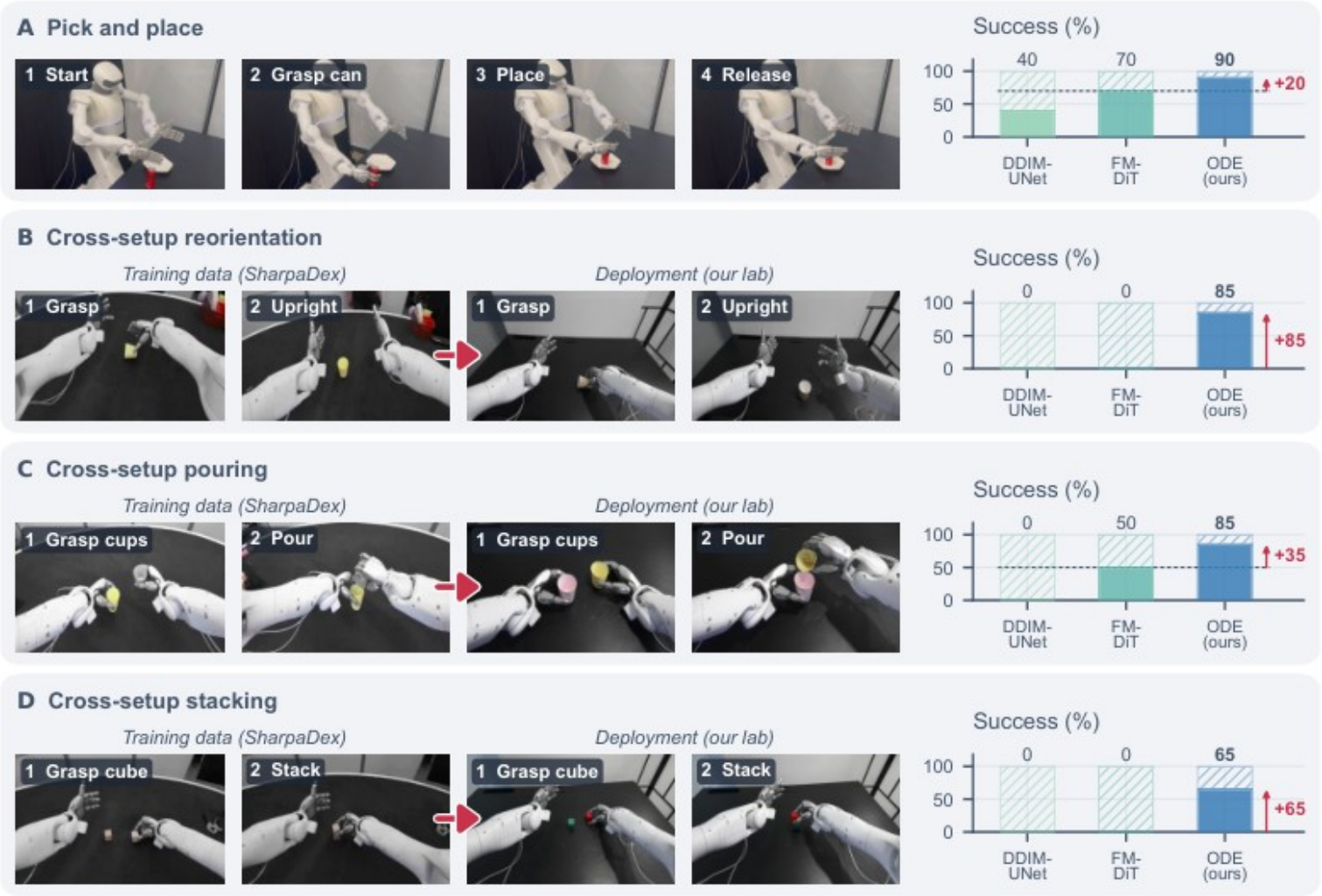


Success rate (%, 20 trials)

- Single-arm Franka: 40 demos, only 50 epochs; block insertion 70 vs 20
- Bimanual Galaxea: microwave with only 34 demos, 75 vs 20
- Dexterous Sharpa: * = trained only on another lab's public data, deployed in our lab
- Humanoid G1: replaces GR00T N1.7 action head, +25 / +40

Note: Some gaps are small (e.g., 10 vs 5, 30 vs 20); sample sizes are limited.

Cross-lab: trained only on others' public data, dexterous skills transfer



Three cross-lab tasks (%)

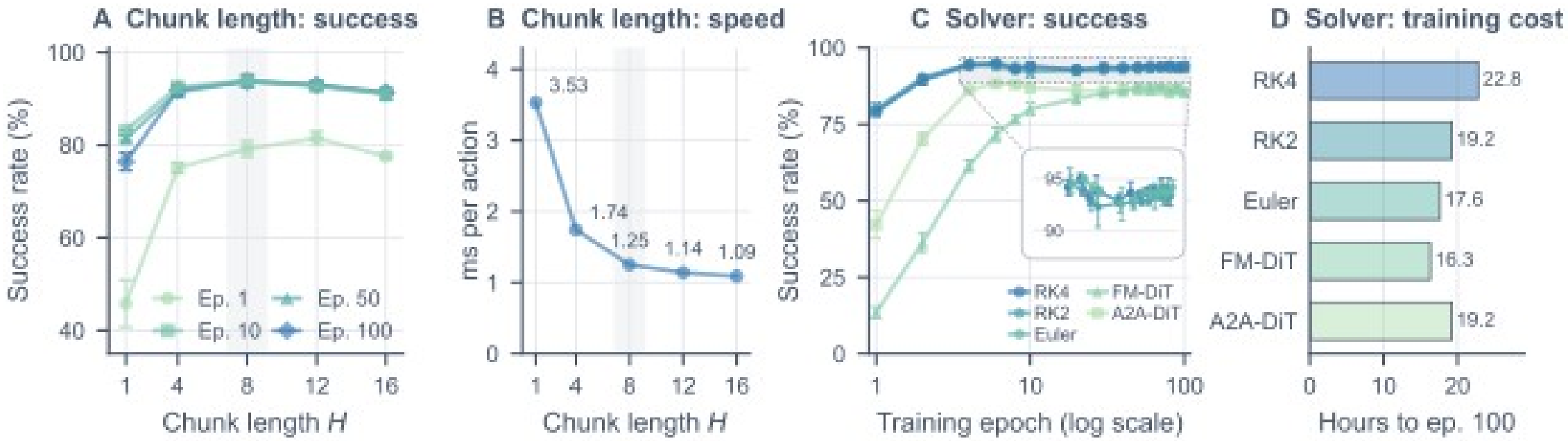
85 — **Upright cup**
FM-DiT / DDIM: 0 / 0

85 — **Bimanual pill pour**
FM-DiT / DDIM: 50 / 0

65 — **Stack blocks**
FM-DiT / DDIM: 0 / 0

- New cups and backgrounds; wooden → colored blocks
- Cup task boundary/interior accel. ratio 0.86 (demos 0.96, baselines 2.03 / 2.97)
- Can grasp (50 local demos): 90 vs FM-DiT 70

Ablations: gains from ODE form, not exact solver or big net



Chunk length

- H = 1: 76.4%, slowest per action
- H ⩾ 4: > 91%; H = 8 best
- H ⩾ 12: occasional loss spikes

Integrator

- RK4 / RK2 / Euler within 1 pt
- Euler train time ≈ FM-DiT
- → Gains not from exact solving

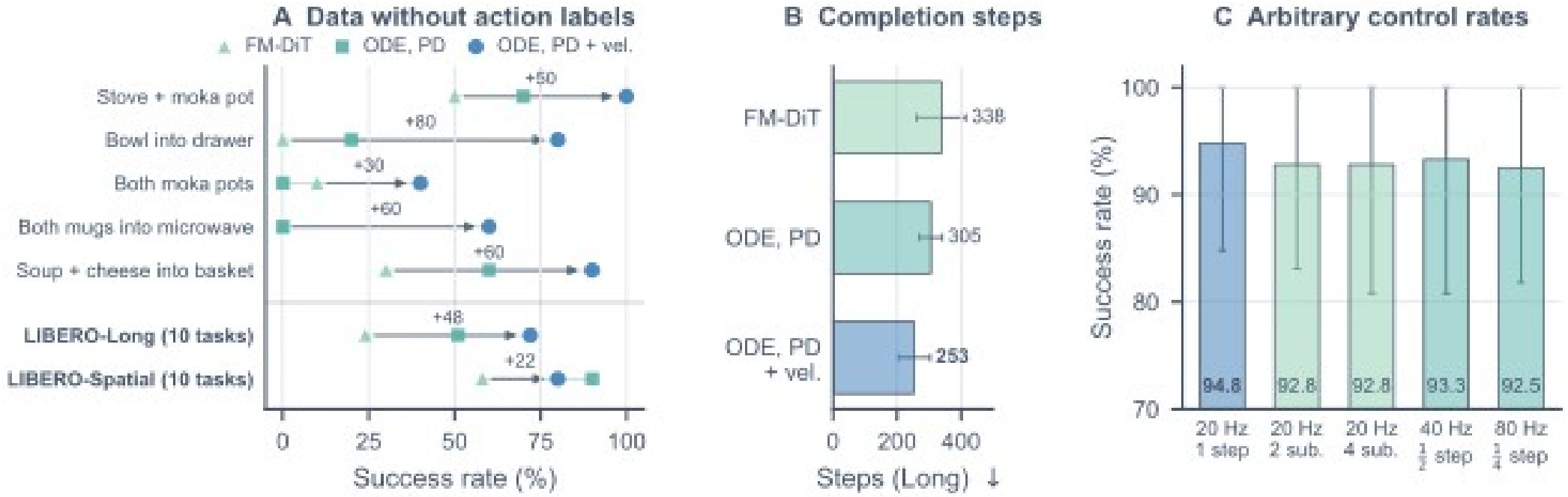
Field architecture

- FiLM-MLP 93.8% vs Transformer 91.2%
- 11.0 ms vs 51.9 ms per chunk
- Integration handles temporal structure

Diversity head

- K need only exceed # modes (default 12)
- No offset: epoch 1 85.3%; mean 79.8%
- Gap ⩽ 4.2 pts from epoch 5

Continuous time: action-free data, any control rate



Action-free: velocity-FF PD

LIBERO-Long **51% → 72%** （FM-DiT 24%）

- Steps per success 305 → 253
- Cost: Spatial 90% → 80%

Any control rate

20 Hz **94.8%** → 40 / 80 Hz **93.3% / 92.5%**

- Change only deploy step size; no retraining
- Sub-step path deviation ⩽ 1 mm

Weakness: discrete signals

- Binary gripper: own classifier head
- Gripper in ODE: ep. 1 79.2%→74.5%, mostly Long
- Model-free RL: gains small, can hurt

Our Assessment

Simple, general paradigm; edge is mainly faster learning

Strengths

- Changes the paradigm, not model size: FiLM-MLP + short-chunk integration; no velocity labels or adjoint
- General: drop-in for visuomotor policies, π0.5 VLA, ImageWAM, GR00T N1.7, SuperMimic RL actor
- "Field takes current action as input" enables contractive perturbation, seamless chunking, streaming
- Real execution gains: fast inference, no chunk-boundary jumps, jerk ⩽ human demos, adjustable control rate
- Broad real-robot coverage, incl. cross-lab transfer and humanoid loco-manipulation

Limitations & Caveats

- Gains are early: abstract's 5.9× / 5.4× / 3.9× are epoch-1 ratios; gaps narrow with full training
- Unequal comparisons: π0.5 head ~3% params, single seed; single-arm real only 50 epochs, hurting diffusion baselines
- 20 real trials/task; some gaps small (10 vs 5, 30 vs 20)
- Limited multimodality: GMM offsets; navigation balance below FM-DiT at epoch 10
- Weaknesses: discrete gripper needs own head; small model-free RL gains; camera shifts all fail; RL needs bounds against drift

Takeaway

Build motion continuity into the policy structure instead of learning it from data

Learns fast

LIBERO epoch 1: 79.2%; 20 demos beat FM with 50

Runs fast

8-step chunk 5.7 ms, ~1/10.8 of FM-DiT; streamable

Moves smoothly

No chunk-boundary jumps; real-robot jerk ⩽ human demos

Applies widely

Plug-and-play for visuomotor / VLA / WAM / humanoid / RL actor

Open questions: Do gains persist with large-scale pretraining and full training? How can discrete events and strong multimodality fit more naturally into the ODE?